

Multimodal data integration study

☐ Not concerned
☐ To do
☐ In progress
☐ Done

PROJECT SETUP

Human Resources

Project manager
Clinician/Biologist
Biostatistician/Bioinformatician/Computational Biologist
Data steward
Ethicist/Legal expert

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Computer Resources

HPC/GPU access
Short-term storage
Long-term storage
Energy footprint assessed

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Good practices

FAIR principles
CARE principles
Open code/data policy defined
Licences

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INTEGRATION PROTOCOL

RESEARCH QUESTION

Formulation

Precise biological/clinical question
Outcomes and measures specified
Success criteria defined

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Scope

Population/system defined
Rationale for multimodality
Exploratory or inferential objectives
Interpretability requirement
Sample size/Power/Biological replicates

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DATA PREPARATION

Governance

Regulatory compliance and re-identification risk
Access, authorisations and consent
Metadata documented
Data Management Plan

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Per-modality preparation

Modalities to collect and/or reuse defined
QC and preprocessing documented
Confounders characterised (batch, bias ...)
Exploratory analysis

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Cross-modal connections

Connection type (vertical/horizontal ...)
Missing modality pattern
Redundancy/complementarity
Data harmonisation

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ALGORITHM SELECTION

Task formulation

Task type defined
Learning strategy chosen
Algorithm assumptions checked against data

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Multimodal integration

Integration timing (early/intermediate/late)
Intermediate fusion approach (if applicable)
Prior resources incorporated

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Practical constraints

Interpretability approach defined
Computational feasibility
Software availability and maintenance

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EVALUATION

Validation

Internal validation strategy
External validation and representativeness
Data leakage risks controlled

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Metrics

Metrics suited to task
Unimodal baseline established
Clinical utility considered

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Multimodal evaluation

Modality ablation and dropout
Cross-modal validation
Modality importance

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Reproducibility

Version control (data, script ...)
Workflow manager
Documentation
Benchmark/data challenge
Pre-registration

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